# Chenfei Xu

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## Profile

Ph.D. candidate at The University of Osaka (expected September 2027), specializing in **multimodal generative modeling for robotics**, including **world models, world-action models**, and cross-modal representation learning, with a focus on **human behavior prediction and future anticipation** in human-robot interaction (HRI). Hands-on experience spanning **multimodal data collection, data processing pipelines, and generative model development** (Video Diffusion Transformers, WMs, WAMs), as well as **embedded and ROS-based robotic system development** and deploying deep learning models on physical robots.
**Fields**: Multimodal Perception & Generation, World-Action Modeling, Robotic System Integration & Deployment, HRI

## Education

**The University of Osaka** — Osaka, Japan
**Ph.D.** in Engineering Science, ***International Honors Students*** — Oct. 2024 – EST. Sep. 2027
Advisors: Prof. Yutaka Nakamura and Prof. Hiroshi Ishiguro, Intelligent Robotics Laboratory

**M.Eng.** in Engineering Science, ***Humanware Innovation Program*** — Oct. 2022 – Sep. 2024
Advisors: Prof. Yutaka Nakamura and Prof. Hiroshi Ishiguro, Intelligent Robotics Laboratory

**Tongji University** — Shanghai, China
**B.Eng.** in Mechanical Engineering, **GPA: 4.59/5.00 (90.92/100)** — Sep. 2018 – Jul. 2022
Advisor: Prof. Qirong Tang, Laboratory of Robotics and Multibody System

## Experience

**Junior Research Associate (JRA)** — Apr. 2025 – Present
*Behavior Learning Team, Information R&D and Strategy Headquarters, RIKEN* — *Kyoto, Japan*

- Designed and implemented **EgoSAS**, a shared egocentric multimodal sensing system for humans and robots, built on **ROS/ROS2 and Raspberry Pi**, integrating fisheye cameras, microphone arrays, IMUs, visual odometry sensors, and eye trackers. [IEICE-CQ 2025]
- Collected and processed **over 60 hours of real-world multimodal interaction data** across diverse scenarios, including daily meetings, robot teleoperation, and visitor reception at *Expo 2025 Osaka*.
- Developed a **multimodal world-action model** (unified-stream diffusion-based joint WAM) to predict multiple plausible future observations and actions in human-populated environments, and integrated it with the humanoid robot platform *"Indy"*.
- Deployed the EgoSAS system Through a Whole-Body Android Avatar, *"Yui"*, to link Avatar-Centric Observations and Operator-Centric Signals. [Humanoids 2026]
- Co-authored two research papers published in the Proceedings of the Annual Conference of JSAI and the proceedings of an **IEEE ICRA 2026 workshop**.

**Research Intern** — Apr. 2023 – Mar. 2025
*Behavior Learning Team, Information R&D and Strategy Headquarters, RIKEN* — *Kyoto, Japan*

- Conducted a proof-of-concept (PoC) study on **action-conditioned future observation generation** and consequence-aware prediction using a masked VAEAC framework. [Artificial Life and Robotics, 2026]
- Developed a deep generative framework for jointly predicting **multiple plausible future** observations and robot actions based on **Video Diffusion Transformers (VDTs)** and **Denoising Diffusion Implicit Models (DDIM)**. [IEEE Access, 2025]
- Deployed a multimodal perception system, deep generative models, and SLAM-based navigation on the mobile robot platform, *MobileTeleco*.
- Orally presented research findings at IEICE Neural Computing Conference 2023, JSME annual Conference on Robotics and Mechatronics (Robomec) 2024, AROB-ISBC 2025 and the Late Breaking Results session of **IEEE/RSJ IROS 2024**

## Projects

**EgoSAS-to-SiWAM: Multimodal World-Action Modeling for HRI** | Webpage 2025 – Present

- Developed **EgoSAS**, a unified egocentric sensing and data processing framework for collecting and aligning multimodal human-robot interaction data across human and robotic platforms.
- Designed **multimodal data processing pipelines** to transform raw video, gaze, audio, and motion signals into unified representations, including semantic segmentation, optical flow, gaze heatmaps, and acoustic maps.
- Developed **SiWAM**, a diffusion-based multimodal world-action model using a **Unified Diffusion Transformer (UDT)** to jointly generate multiple plausible future observations, gaze, acoustic cues, and robot actions.
- Implemented **modality-aware masking and multi-task training** to leverage datasets with partially missing modalities, and evaluated the model on human gaze prediction and anticipation of occluded pedestrians in real-world HRI scenarios.

***Tech Stack:*** *PyTorch, WMs/WAMs, OpenCV, YOLO, Multimodal Sensor Fusion (Optical Flow/Gaze/Acoustic Mapping)*

**Multi-Future Prediction for Mobile Robots** | GitHub | Paper 2023 – 2025

- Developed a **diffusion-based generative framework** for jointly predicting future visual observations and robot actions in human-populated environments.
- Implemented **Video Diffusion Transformers (VDTs)** with DDIM sampling to generate multiple plausible future observation-action trajectories from the same observed context.
- Integrated the generative framework into the **Mobile Teleco** robotic platform as a foundational module for predicting candidate future observations and actions in real-world environments.

***Tech Stack:*** *PyTorch, ViT/DiT/VDT, MaskDINO, DDIM, ROS/ROS2, Raspberry Pi, Serial Communication, 2D SLAM*

## Selected Publications

[6] **Multimodal Ego-Centric Sensing for Social Interaction World Action Models in Real-World HRI**
**Chenfei Xu**, Zhichao Chen, Huthaifa Ahmad, Liliana Villamar Gómez, Yuya Okadome, Hiroshi Ishiguro, Yutaka Nakamura
*Under review, submitted to IEEE Robotics & Automation Magazine, Sep. 2026.*

[5] **Action-Inclusive Multi-Future Prediction Using a Generative Model in Human-Related Scenes for Mobile Robots**
**Chenfei Xu**, Huthaifa Ahmad, Yuya Okadome, Hiroshi Ishiguro, Yutaka Nakamura
*IEEE Access, vol. 13, pp. 167034-167044, 2025.*

[4] **A Situation-Dependent Motion Generation Framework Using a Generative Model of Motion and Observation**
**Chenfei Xu**, Yuya Okadome, Hiroshi Ishiguro, Yutaka Nakamura
*Artificial Life and Robotics, 2026.*

[3] **An Ego-centric Situational Awareness Sensor (EgoSAS): Collecting Human-Human Interaction Data for Mobile Robots' Co-existence**
**Chenfei Xu**, Zhichao Chen, Yuya Okadome, Hiroshi Ishiguro, Yutaka Nakamura
*IEICE Technical Committee on Communication Quality (IEICE-CQ), 2025.*

[2] **A Social-interaction World Model Pipeline: Multimodal Data Acquisition for Capturing Bifurcating Social Intents in HRI**
Yutaka Nakamura, **Chenfei Xu**, Yuma Kabe, Kaoruko Shinkawa, Yuya Okadome, Yoshihiro Nakata
*ICRA 2026 Workshop: From Data to Decisions: VLA Pipelines for Real Robots, 2026.*

[1] **Exploring Implicit Gestures in Egocentric Daily Activities for Robot Motion Design**
Liliana Villamar Gómez, **Chenfei Xu**, Huthaifa Ahmad, Yutaka Nakamura
*The 40th Annual Conference of the Japanese Society for Artificial Intelligence (JSAI), 2026.*

## Skills

**Languages:** Mandarin (Native), English (Fluent, TOEFL iBT 99), Japanese (Fluent, JLPT N1)
**Robotic Platforms:** Mobile Teleco (mobile robot), Yui and Indy (humanoid robots)
**Technical Skills:** Python, C/C++ (Working Knowledge); PyTorch, Hugging Face Transformers, Diffusers; ROS/ROS2, WebRTC, UDP, Serial Communication; Git, Docker, Linux, Raspberry Pi OS

## Awards

**Humanware Innovation Doctoral Program** Apr. 2023 – Present
*The University of Osaka*

**International Honors Students with Excellent academic standing** 2024 – 2027
*The University of Osaka* *Full tuition exemption (Doctoral course)*